

Recovery Tools

GetDataBack

www.runtime.org

GetDataBack helps recover data when your drive's partition table, boot record, FAT or root directory is corrupt or lost due to viruses, power failure or by accidental deletion. It may even recover data when the drive is no longer recognised by the OS or when all directory information is lost. Its main advantage is that it runs in a read-only mode, which means the program will never attempt to write to the drive you are about to recover.

Active Partition Recovery

www.partition-recovery.com

Active Partition Recovery is a DOS tool that helps recover deleted partitions and logical drives by restoring the information from a backup of the Master Boot Record, partition tables and boot sectors.

Stellar Phoenix Data Recovery Software

www.stellarinfo.com

Stellar Phoenix analyses and recovers data from inaccessible disks that have been corrupted by virus attacks and accidental formats. It supports FAT16 and FAT32 file systems and has been known to have high success rates.

Zero Assumption Recovery

www.z-a-recovery.com

This tool extracts data from damaged hard disks. It works from within Windows and can recover files from FAT16, FAT32 and NTFS partitions. It supports long filenames and NTFS compression.

PC Inspector File Recovery

www.convar.com

This data recovery program supports the FAT12, FAT16, FAT32 and NTFS file systems. It's one of the few free file recovery software.

FILERECOVERY for Windows

www.lc-tech.com

This software can undelete accidentally deleted files from various file systems under Windows. It can be run from a floppy to avoid overwriting any files on the hard disk.

Hard disks store data in sectors and clusters. Think of sectors and clusters as a hotel with floors and rooms. The guest list is stored at the reception and any one at the desk is able to tell you the room in which a particular guest is staying. Hard disks store information about each file in a similar way. The 'address' of each file is stored in a separate area of the hard disk that is not accessible with common tools. When a file is requested, the disk looks up this 'register', locates the file and passes it on. On FAT and FAT32 partitions, this entry is stored in the FAT (File Allocation Table).

When a file is 'permanently' deleted, only its entry from this 'register' is removed. The actual file is still stored on the disk until some program requires space to write to. Thus, it may still be possible to recover the file.

The sooner you realise that a required file has been deleted, the better. The further you need to go back in time, the more luck you need. Basically, the more the hard disk is used, the less the chances of recovery, since the probability of the clusters being overwritten with fresh data is higher.

The nature of the applications you use determines how the application modifies the registry or uses dependant files. A browser, for example, is the worst enemy of data recovery—it saves hundreds of files as cache, tremendously increasing the probability of clusters being overwritten. E-mail is a much lesser cause for fear—as you download e-mail, only a few kilobytes would be used up every time. A word processor doesn't write to any dependant files, but if any application settings are changed, it would modify the registry. If the registry size increases because of this, it may look for a free cluster to write to.

Another criteria for higher chances of successful recovery is the location of your stored data, again amounting to how the hard disk is used. For example, if all your applications and Windows are on drive C,

the Windows swap file is also on the C: drive and all your data (including e-mail, browser favourites, etc.) is on drive C. Your chances of recovery would be the highest because no program is transparently moving data to drive D.

Recovering lost files

The minute you realise that you want to recover a file, stop using the PC—close any programs, shut down Windows, hit [Ctrl] + [S] or even chat. You don't want any application to write anything to the hard disk. Then, take a step back and think out exactly how you want to go about recovering the file. Usually, there's room for experimenting. Also, when recovering files, save them on a drive other than the one you are recovering from. If you don't want any data to be written to the drive until you finish recovering,

[illegible]

*File Rescue displays a list of deleted files
their condition*

There are several tools available that can help you recover the lost file(s). One of them is **Rescue (www.softwareself.com)** is a small and useful tool that works with FAT, NTFS, and NTFS drives. When you launch the program, it asks you for the drives to scan and what kind of files to look for. You select your selections and click OK. It scans the disk and shows up a list of files to recover.



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8



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